

Somanshu Mohapatra

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SKILLS

Languages: C++, Python, Embedded C, MATLAB (Simulink, Robotics Toolbox).

Robotics: Forward/Inverse Kinematics (DH Parameters), ROS2, Gazebo, Path Planning, Motion Control, PID Control, Sensor Fusion, SLAM.

Electronics: Arduino UNO Q, Raspberry Pi, STM32, ATmega328P, RP2040, Motor Drivers (DRV8825, TB6600), Serial Communication (UART, I2C, SPI, CAN).

CAD & Design: SolidWorks, Fusion 360, AutoCAD, Creo Parametric, FEA, Assembly Design, Rapid Prototyping, 3D Printing (FDM/SLA).

Soft Skills: Effective Organizer, Team Management, Strong Interpersonal Skills, Problem Solving.

INTERNSHIP

AutomateX – PLC based training for Industrial Automation

Jun' 25 – Jul' 25

Student Intern | Ladder Logic, LogixPro, TwidoSuite

- Engineered an automated feeder control system in LogixPro using ladder logic, reducing manual labor by 70% and feed wastage by 25%.
- Implemented safety interlocks, emergency-stop recovery, and auto/manual modes to ensure zero-failure restart behavior and reliable operation.

PROJECTS

Dynamic Audio Monitoring and Narration | Raspberry Pi, Python, AssemblyAI API

Apr' 25 – May' 25

- Developed a Python-based real-time transcription system for the deaf using Raspberry Pi 4B, achieving 95%+ accuracy with <300ms latency.
- Integrated AssemblyAI WebSockets with an I2C SSD1306 OLED display using multi-threading to ensure seamless, non-blocking subtitle updates.

3-DOF Robotic Arm with Real-Time Control | MATLAB, SolidWorks, Arduino IDE, Bambu Studio

Apr' 24

- Constructed a 3-DOF robotic arm using NEMA 17 steppers and Arduino Mega, utilizing 3D-printed components to optimize payload-to-weight ratio.
- Implemented forward and inverse kinematics via MATLAB-UART communication, reducing trajectory error by 20% during real-time tasks.

Smart Energy Meter | ESP32, IoT Monitoring, HTML Webserver

Jun' 23

- Designed an IoT-based energy monitoring system using ESP32 and Wi-Fi to track power consumption via a real-time browser interface.
- Integrated ZMPT101B and ZMCT103C sensors to measure voltage and current with $\pm 3\%$ accuracy visualized via custom HTML dashboard.

CERTIFICATES

Data Base Management System | NPTEL

Aug' 25

AutomateX: Mastering PLCs for Industrial Automation | LPU

Mar' 25

Fusioncraft | Resolute Lab

Mar' 24

ACHIEVEMENTS

METHOD FOR ENHANCING FOOD ACCESSIBILITY, MONITORING FRESHNESS, AND REDUCING ENERGY LOSS IN CYLINDRICAL REFRIGERATION UNITS

RE-Sep' 25

- Somanshu Mohapatra, Dr. Sumit Shoor
- Application No.: 202511085342 A

EDUCATION

Lovely Professional University <i>Bachelor of Technology in Robotics and Automation Engineering — CGPA: 8.38</i>	Phagwara, Punjab <i>Aug' 23 – Present</i>
Shri S.N. Siddeshwar Public School <i>Intermediate, PCMB — Percentage: 85.6%</i>	Gurugram, Haryana <i>Mar' 21 – May' 23</i>
Ryan International School <i>Matriculation — Percentage: 81.6%</i>	Gurugram, Haryana <i>Mar' 20 – May' 21</i>